Keyu He

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Education

Carnegie Mellon University, Pittsburgh, PA **May 2027**
M.S. in Intelligent Information Systems, School of Computer Science **GPA: 4.14/4.33**

University of Southern California, Los Angeles, CA **May 2025**
B.S. in Computer Science; B.A. in Applied & Computational Mathematics **GPA: 3.98/4.00**
Minor in Artificial Intelligence Applications

Research Interests

Socially intelligent and multi-agent AI, verifiable-reward environments and reinforcement learning, LLM evaluation, human–AI interaction.

Skills

Programming: Python, C++, TypeScript/JavaScript, Java, SQL, C, x86-64 Assembly
ML/Research: PyTorch, NumPy/Pandas, LLM/VLM evaluation, multi-agent systems, reinforcement learning
Systems/Tools: Databricks, AWS, RAG, LangChain, FAISS, Git, VS Code extensions, REST APIs, LaTeX

Publications

1. **Keyu He**, Xuhui Zhou, Maarten Sap. *Social Gym and SPaRTan: Benchmarking and Improving LLM Social Reasoning via Multi-Agent Game Tournaments.* Preprint, 2026. 10.48550/arXiv.2608.09128
   [rule-verifiable multi-agent environments; LLM orchestration; large-scale benchmarking; self-play design]
2. **Keyu He**, Tejas Srinivasan, Brihi Joshi, Xiang Ren, Jesse Thomason, Swabha Swayamdipta. *Believing without Seeing: Quality Scores for Contextualizing Vision-Language Model Explanations.* ACL 2026. 10.48550/arXiv.2509.25844
   [VLM metric design; calibration; human-subject studies; statistical analysis]
3. **Keyu He***, Qianou Ma*, Valerie Chen, Wayne Chi, Tongshuang Wu. *RECAP: An End-to-End Platform for Capturing, Replaying, and Analyzing AI-Assisted Programming Interactions.* ACL Demo 2026. 10.48550/arXiv.2605.01104
   [VS Code instrumentation; full-stack TypeScript development; cloud data pipelines; human–AI interaction analysis]
4. Brihi Joshi*, **Keyu He***, Sahana Ramnath, Sadra Sabouri, Kaitlyn Zhou, Souti Chattopadhyay, Swabha Swayamdipta, Xiang Ren. *ELI-Why: Evaluating the Pedagogical Utility of LLM Explanations.* Findings of ACL 2025. 10.48550/arXiv.2506.14200
   [benchmark design; human-subject studies; web-app implementation; multi-metric LLM evaluation]
5. Huihan Li*, Arnav Goel*, **Keyu He**, and Xiang Ren. *Attributing Culture-Conditioned Generations to Pretraining Corpora.* ICLR 2025. 10.48550/arXiv.2412.20760.
   [large-scale corpus analysis; Python data engineering and visualization; survey server infrastructure design]

Research Experience

Social Gym and SPaRTan 2026–Present
Carnegie Mellon University; with Prof. Maarten Sap

- Built **Social Gym**, a Python, game-agnostic N-agent environment with finite-state game logic, partial observability, and rule-defined rewards across 21 games, enabling reproducible multi-turn evaluation without LLM judges.
- Ran role-balanced tournaments for seven closed- and open-weight LLMs across 17 competitive and four cooperative games, covering 560+ episodes per model; fit Bradley–Terry/Elo ratings and role-conditioned capability profiles.
- Designed **SPaRTan**, a training-free play–reflect–transfer loop and evaluated within-game iteration, cross-game and multigame transfer, and distillation into six student models.
- Extending the environment toward RLVR by generating social scenarios with rule-computed outcomes.

Believing without Seeing 2024–2026
USC DILL Lab; with Profs. Swabha Swayamdipta and Xiang Ren

- Designed Visual Fidelity and Contrastiveness, two quality scores for VLM explanations, and evaluated their calibration against model correctness on A-OKVQA, VizWiz, and MMMU-Pro.
- Designed and ran Prolific studies with 300 annotations per dataset and condition; the combined scores improved users' correctness judgments by 11.1 percentage points and reduced false belief in incorrect predictions by 15.4 points.

RECAP 2025–2026
Carnegie Mellon University; with Prof. Tongshuang Wu

- Co-built an open-source VS Code instrumentation and replay platform that captures Copilot chats and fine-grained shadow-git edits, links AI suggestions to code, and supports extensible behavior analysis.
- Built the TypeScript extension and cloud data pipeline for a deployment spanning 41 students and 406 work sessions, capturing 2,034 prompts and 8,239 code edits.

ELI-Why 2024–2025
USC INK and DILL Labs; with Profs. Xiang Ren and Swabha Swayamdipta

- Co-created a 13.4K-question benchmark for testing whether LLM explanations adapt to elementary, high-school, and graduate-level learners; led task, rubric, and prompt design and built the human-study web application.
- Ran two human studies across 811 participants and multi-metric analyses, showing GPT-4 explanations matched their requested grade level 50% of the time versus 79% for human explanations.

Attributing Culture-Conditioned Generations to Pretraining Corpora 2024
USC INK Lab; with Prof. Xiang Ren

- Contributed large-scale corpus analysis, Python data engineering, and visualization to MEMOed, a framework for attributing culture-conditioned generations across 110 cultures.
- Built Google APIs survey infrastructure and a routing service that directed participants to culture-specific forms for scalable data collection.

Experience

Adobe, San Jose, CA May 2026 – Aug. 2026
Machine Learning Engineer Intern

- Shipped 5+ agent skills, including a four-skill family outside the team's own domain, through an **agent skill generator** merged into the team's shared skill repo, which turns rough developer requests into standards-compliant skills, grounding against internal prior art so existing work isn't rebuilt and self-auditing against the agentskills.io spec.
- Developed the systematic quality measure for the team's agent recommendation pipeline, a **reference-free 6-dimension rubric** combining LLM-as-judge with deterministic checks, reproducible to 0.01–0.05 per-dimension variance across repeated runs and surfacing two systematic weaknesses for the team to act on.
- Took that measure to production across the org: ported it onto the internal GenAI evaluation service as a reusable scorer and staged it end-to-end for two enterprise customers, on a four-stage Databricks pipeline built to run it.

CMind, Remote May 2025 – Aug. 2025
Data Engineer Intern

- Built a LangChain-based RAG prototype over an internal analytics DB (chunking, embeddings, FAISS indexing/retrieval, LLM answer synthesis) with metadata filters and notebook/API utilities.
- Designed CEO-speech trait metrics and an auto-evaluation pipeline with custom rubrics, achieving Fleiss' κ of 0.89–0.91 against human annotations.

Teaching Experience

Carnegie Mellon University, Language Technologies Institute Aug. 2026 – Present

Graduate Teaching Assistant, 11-711 Advanced Natural Language Processing

- Supporting a graduate NLP research course covering modeling and learning algorithms and culminating in an original research project.

University of Southern California, Los Angeles, CA Aug. 2022 – May 2025

Teaching & Grading Assistant

- Coordinated logistics and grading for two CS courses (CSCI-102, CSCI-360), serving ∼300 students per term.
- Led weekly OHs/discussions for ∼20 students, clarifying concepts and providing feedback on assignments.

Projects

VLM Hallucination Mitigation — Carnegie Mellon University Feb. 2026 – Apr. 2026

- Built a training-free latent reflection module that monitors decoding entropy and injects reflection steps when the model is uncertain.
- Benchmarked against DAMO, VCD, OPERA, and DeGF on POPE/RePOPE/MMBench across LLaVA-1.5-7B, InstructBLIP-7B, and Qwen2-VL-7B.
- On RePOPE, raised LLaVA-1.5-7B accuracy from 84.0% to 90.3% with the lowest calibration error (ECE 0.028) among all methods compared.

LLM Prompt Recovery Project — University of Southern California Mar. 2024 – Apr. 2024

- Built a system to reconstruct user prompts from original text and Gemma-modified outputs.
- Fine-tuned Mixtral-8x7B with custom similarity metrics (sentence-T5-base + sharpened cosine), achieving a 0.65 leaderboard score.
- Earned a Kaggle silver medal and released the final fine-tuned model publicly on Kaggle.

Major Awards

- **USC CURVE Research Fellowship**, awarded in 4 terms (S24, Su24, F24, S25); total funding **$6,750**
- **USC Academic Achievement Award**, awarded in 4 terms (F22, S23, S24, F24); total support ≈**$24,000**
- **Silver Medal**, Kaggle Competition, Ranked 75/2175 (Top 3.4%) on the global leaderboard, LLM-Prompt-Recovery Project, 2024
- **1st Prize**, International Linguistics Olympiad (Senior Level), Individual Open Round, China, 2021
- **1st Prize**, International Linguistics Olympiad (Senior Level), Team Open Round, China, 2021
- **4th Place**, USC Integral Bee Competition, 2022